

Alab-mart

Alab-mart is a ecommerce company which is completely working online through a ecommerce website we are partnered with different book stores and hardware companies to sell the books ,hardware devices related to AI. Basically in this platform only ai related stuff will be available like books,hardware,ai assistants .

The owner of this ecommerce platform is A.Kalyan Sai,He is an ai engineer with 2 internship experiences one at kodemelon technologies and other one is at celtn there he learnt about cutting edge ai technologies like rag ,computer vision, automation pipelines.

In this mart the offers will be on 3 days per year
that is on 09-09,26-11 and 3-11 of every year

If you order any item or bluck of items then all the items will be delivered within 2 days of your purchase

For any queries contact details are as follows
for ecommerce website queries

contact no: 1234567890

email: ecommerce@alab-mart.com

for product enquires
contact no : 9876543210

email:product@alab-mart.com

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ABOUT ALAB-MART

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ALAB-MART is an online e-commerce platform dedicated exclusively to Artificial Intelligence products. Our mission is to provide students, developers, researchers, and AI professionals with a trusted marketplace where they can purchase AI books, AI hardware devices, and AI-powered smart assistant devices.

We partner with leading publishers, electronics manufacturers, and technology companies to ensure customers receive genuine products at competitive prices.

Mission:

To make Artificial Intelligence learning and development resources easily accessible to everyone.

Vision:

To become India's leading AI-focused e-commerce platform.

Head Office:

Guntur, Andhra Pradesh India

Business Model:

Business-to-Customer (B2C)

Delivery Coverage:

Pan India

Customer Support:

Monday to Friday

9:00 AM to 8:00 PM IST

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SHIPPING POLICY

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- Orders are processed within 24 hours.
- Standard delivery takes 2 business days.
- Orders above ₹5000 qualify for free shipping.

- Customers receive a tracking ID once the order is shipped.
- Orders are delivered Monday to Saturday.
- Delivery may be delayed during festivals or natural disasters.
- Customers should inspect packages before accepting delivery.
- Damaged shipments should be reported within 24 hours.

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RETURN POLICY

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Products may be returned within 7 days after delivery.

Books must be unused.

Hardware products must be returned with original accessories.

AI Assistant devices must be factory reset before returning.

Products physically damaged by customers cannot be returned.

Refunds are processed only after inspection.

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REFUND POLICY

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Refunds are initiated within 2 business days after approval.

The refund is credited to the original payment method.

Customers receive call notifications during every refund stage.

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WARRANTY POLICY

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Books:

Manufacturing defects only.

AI Hardware:

One-year manufacturer warranty.

AI Smart Assistants:

One-year manufacturer warranty.

Warranty does not cover accidental damage, water damage, or unauthorized repairs.

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PAYMENT METHODS

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EMI for eligible products

Cash on Delivery is currently available.

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GENERAL FAQ

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Q: How long does delivery take?

A: Usually within 2 business days.

Q: Do you deliver across India?

A: Yes.

Q: Can I cancel my order?

A: Yes, before shipment.

Q: Can I track my order?

A: Yes.

Q: How do I contact customer support?

A: Using the phone numbers or email provided.

Q: Which payment methods are accepted?

A: UPI, Cards, Wallets and Net Banking.

Q: Do products have warranty?

A: Hardware and Smart Assistants include manufacturer warranty.

Q: Are all products genuine?

A: Yes.

Q: Can I buy multiple quantities?

A: Yes.

Q: Are prices inclusive of GST?

A: Yes.

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BUYING GUIDE

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For Beginners:

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

For Deep Learning:

Deep Learning by Ian Goodfellow

For Complete AI:

Artificial Intelligence: A Modern Approach

For MLOps:

Designing Machine Learning Systems

For NLP:

Natural Language Processing with Transformers

For Edge AI:

Jetson Orin Nano

For Robotics:

Jetson AGX Orin

For IoT:

Raspberry Pi 5

For Low-Power AI:

Google Coral Dev Board

For Smart Home:

Apple HomePod

For Budget Smart Speaker:

Amazon Echo Dot

For Android Users:

Google Nest Audio

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POPULAR COMPARISONS

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Jetson Orin Nano vs Raspberry Pi 5

Jetson Orin Nano offers significantly higher AI performance.

Raspberry Pi 5 is better suited for education and IoT projects.

Apple HomePod provides better sound quality.

Amazon Echo integrates more smart home devices.

Google Nest Audio is ideal for Android users.

Hands-On Machine Learning is best for practical coding.

Deep Learning is more theory-focused.

Artificial Intelligence: A Modern Approach provides the most comprehensive AI coverage.

Alab-mart consists of 3 different products they are as follows:

1. AI Books
2. AI Hardware devices
3. AI Assistant devices

The items we offer in AI Books product is

Here is the expanded, highly detailed breakdown for all 10 books following your requested format:

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

- **Product ID:** 1
- **Product Name:** Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (3rd Edition)
- **Brand:** O'Reilly Media
- **Category:** Computer Science / Artificial Intelligence / Machine Learning
- **Price:** ~\$69.99 USD (List Price) / ₹3,250 – ₹8,729 INR (Print editions depending on binding/imprint)
- **Availability:** In Stock across major retailers (Amazon, O'Reilly Learning Platform, Barnes & Noble).
- **Rating:** 4.8 / 5 stars (from 2,500+ global ratings)

- **Short Description:** A practical, code-first guide using minimal theory to teach software developers end-to-end machine learning and deep learning using Python frameworks.
- **Detailed Description:**

This book provides a pragmatic, project-driven approach to mastering machine learning. Author Aurélien Géron walks programmers through building production-ready systems without getting bogged down in excessive mathematical theory. The first half focuses on traditional ML algorithms using **Scikit-Learn**—including classification, regression, SVMs, decision trees, ensemble methods, and unsupervised learning (clustering, PCA). The second half transitions into deep learning using **Keras** and **TensorFlow 2**, covering Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs) for computer vision, Recurrent Neural Networks (RNNs) and LSTMs for sequential data, Transformers, Generative Adversarial Networks (GANs), Diffusion Models, and Deep Reinforcement Learning.

- **Key Features:**
 - End-to-end walkthrough of a real-life California Housing project (data preparation, model evaluation, hyperparameter tuning).
 - Step-by-step guidance on training and scaling deep neural networks across multiple GPUs and TPUs.
 - Practical strategies for dealing with data distribution shifts, missing values, and high dimensionality.
 - Chapter-ending exercises with comprehensive online solutions for active learning.
- **Technical Specifications:**
 - **Author:** Aurélien Géron
 - **Edition:** 3rd Edition (Published Oct 2022)
 - **Page Count:** 834 pages
 - **Language:** English
 - **Tech Stack:** Python 3.10+, Scikit-Learn 1.1+, TensorFlow 2.10+, Keras, Pandas, NumPy, Matplotlib.
- **Package Contents:** 1x Paperback Book or eBook License; full access to the official public GitHub Repository with interactive Jupyter Notebooks.
- **Warranty:** 30-day standard merchant replacement guarantee for physical printing defects or transit damage.
- **Shipping Information:** Dimensions: 7.0 x 1.8 x 9.2 inches; Weight: ~1.42 kg; standard worldwide shipping options available.
- **Frequently Asked Questions:**
 - *Q: Is advanced calculus required?* A: No, basic linear algebra and Python fluency are sufficient.
 - *Q: How does the 3rd edition differ from the 2nd edition?* A: It adds modern Transformers, Diffusion Models, updated Keras APIs, and improved MLOps deployment tools.
- **Customer Reviews:**
 - "The single best entry point into applied machine learning for software developers."
 - "Combines clean code with practical explanations better than any online course."

2. Deep Learning

- **Product ID:** 2
- **Product Name:** Deep Learning (Adaptive Computation and Machine Learning series)
- **Brand:** MIT Press
- **Category:** Deep Learning / Theoretical Computer Science / Applied Mathematics

- **Price:** ~\$80.00 USD / ₹800 – ₹5,999 INR
- **Availability:** In Stock globally in Hardcover and free digital web format.
- **Rating:** 4.7 / 5 stars (from 1,800+ global ratings)
- **Short Description:** Widely recognized as "The Bible of Deep Learning," offering complete theoretical and mathematical foundations of deep neural networks.
- **Detailed Description:**

Authored by field pioneers Ian Goodfellow, Yoshua Bengio, and Aaron Courville, this text provides a rigorous mathematical introduction to the principles underlying deep learning. Divided into three main parts: **Applied Math and Machine Learning Basics** (linear algebra, probability, information theory, numerical computation), **Deep Networks: Modern Practices** (deep feedforward networks, regularization, optimization algorithms, CNNs, sequence modeling), and **Deep Learning Research** (autoencoders, representation learning, generative adversarial networks, Monte Carlo methods, partition functions).

- **Key Features:**
 - Groundwork in linear algebra, probability, numerical computation, and vector calculus.
 - Comprehensive theoretical treatment of loss functions, backpropagation mechanics, and optimization landscapes (SGD, Adam, RMSProp).
 - In-depth theoretical models for generative networks, including GANs (invented by co-author Ian Goodfellow).
- **Technical Specifications:**
 - **Authors:** Ian Goodfellow, Yoshua Bengio, Aaron Courville
 - **Edition:** 1st Edition (2016)
 - **Page Count:** 800 pages
 - **Language:** English
 - **Format:** Hardcover / HTML Web Edition
- **Package Contents:** 1x Hardcover Edition or eBook access code.
- **Warranty:** Retailer return/replacement policy for damaged goods.
- **Shipping Information:** Dimensions: 7.3 x 1.6 x 9.2 inches; Weight: ~1.63 kg.
- **Frequently Asked Questions:**
 - **Q:** *Does this book contain code implementations in Python?* **A:** No, it is code-agnostic and relies on mathematical formulations and pseudocode.
 - **Q:** *Is this suitable for beginners?* **A:** It is best suited for readers with a strong background in college-level linear algebra and multivariate calculus.
- **Customer Reviews:**
 - "An absolute masterpiece for understanding *why* neural networks work, not just how to run them."
 - "Essential reference book for AI researchers and PhD candidates."

3. Artificial Intelligence: A Modern Approach

- **Product ID:** 3
- **Product Name:** Artificial Intelligence: A Modern Approach (4th Edition)
- **Brand:** Pearson
- **Category:** Artificial Intelligence / Computer Science / Academic Textbooks
- **Price:** ~\$120.00 – \$180.00 USD / ₹1,200 – ₹9,500 INR
- **Availability:** In Stock; widely available as a primary university textbook.
- **Rating:** 4.7 / 5 stars (from 1,200+ global ratings)

- **Short Description:** The gold-standard university textbook providing an exhaustive overview of the entire domain of classical and modern AI.
- **Detailed Description:**

Written by Stuart Russell and Peter Norvig, *AIMA* is the standard textbook used in thousands of university AI courses. Built around the unifying concept of an **Intelligent Agent**, it covers classical AI (uninformed/informed search, constraint satisfaction, game playing), knowledge representation and logic, probabilistic reasoning under uncertainty, reinforcement learning, computer vision, natural language processing, robotics, and the societal and ethical impacts of AI. The 4th edition thoroughly integrates modern deep learning and probabilistic programming alongside classical symbolic methods.

- **Key Features:**
 - Unifying "Intelligent Agent" architecture framing all algorithms.
 - Full spectrum coverage: Knowledge Representation, First-Order Logic, Bayesian Networks, Planning, Machine Learning, Deep Learning, and NLP.
 - Expanded sections on ethical AI, safety, privacy, and fairness.
- **Technical Specifications:**
 - **Authors:** Stuart Russell, Peter Norvig
 - **Edition:** 4th Edition (2020)
 - **Page Count:** 1,136 pages
 - **Language:** English
 - **Publisher:** Pearson
- **Package Contents:** 1x Hardcover / Paperback Book; online repository access (AIMA-Python / AIMA-Java).
- **Warranty:** Standard retailer book exchange policy.
- **Shipping Information:** Dimensions: 8.6 x 1.7 x 10.9 inches; Weight: ~2.15 kg.
- **Frequently Asked Questions:**
 - *Q: Is this book purely theoretical?* A: It covers foundational concepts with formal pseudocode, backed by open-source Python implementations on GitHub.
 - *Q: Is it up to date with deep learning?* A: Yes, the 4th edition spent considerable effort updating chapters to cover modern deep neural models alongside symbolic AI.
- **Customer Reviews:**
 - "The definitive encyclopedia of Artificial Intelligence."
 - "Broad, deep, and brilliantly written."

4. Designing Machine Learning Systems

- **Product ID:** 4
- **Product Name:** Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications
- **Brand:** O'Reilly Media
- **Category:** ML Systems Architecture / MLOps / Software Engineering
- **Price:** ~\$59.99 USD / ₹1,368 – ₹1,600 INR
- **Availability:** In Stock.
- **Rating:** 4.8 / 5 stars (from 600+ global ratings)
- **Short Description:** A holistic operational guide for building reliable, scalable, maintainable, and adaptive ML systems in production environments.
- **Detailed Description:**

Chip Huyen bridges the gap between isolated Jupyter notebooks and production-scale enterprise software. This book teaches how to design holistic ML architectures, covering business metrics alignment, data engineering fundamentals (batch vs. stream processing, feature stores), training data collection and labeling, feature engineering, offline model evaluation, model serving strategies, data distribution shifts (covariate shift, concept drift), monitoring, and continual learning pipelines.

- **Key Features:**
 - Detailed breakdown of real-time streaming architectures (Kafka, Flink) vs. batch pipelines.
 - Strategies for detecting and mitigating data drift, concept drift, and schema changes in production.
 - Production deployment techniques: model quantization, pruning, batch vs. online inference, and edge deployment.
 - Case studies on real-world system design interview questions.
- **Technical Specifications:**
 - **Author:** Chip Huyen
 - **Edition:** 1st Edition (2022)
 - **Page Count:** ~386 pages
 - **Language:** English
 - **Focus Stack:** Feature Stores, Data Pipelines, Streaming Infrastructure, MLOps Monitoring Tools.
- **Package Contents:** 1x Paperback Book or Digital Copy.
- **Warranty:** Publisher/Retailer 30-day exchange window.
- **Shipping Information:** Dimensions: 7.0 x 0.75 x 9.0 inches; Weight: ~0.62 kg.
- **Frequently Asked Questions:**
 - *Q: Does this book teach how to write ML algorithms?* A: No, it focuses on the operational infrastructure, system design, and lifecycle around models.
 - *Q: Is this helpful for Machine Learning System Design interviews?* A: Yes, it is widely considered one of the primary preparation books for ML System Design interviews at top tech companies.
- **Customer Reviews:**
 - "Must-read for software engineers moving to ML and MLOps roles."
 - "Fills a huge gap in the industry that traditional ML books ignore."

5. Machine Learning Engineering

- **Product ID:** 5
- **Product Name:** Machine Learning Engineering
- **Brand:** True Positive Inc.
- **Category:** MLOps / Applied Engineering / Best Practices
- **Price:** ~\$35.00 – \$45.00 USD / ₹1,200 – ₹3,500 INR
- **Availability:** In Stock.
- **Rating:** 4.7 / 5 stars (from 450+ global ratings)
- **Short Description:** A concise, direct handbook detailing state-of-the-art engineering practices for executing end-to-end ML projects successfully.
- **Detailed Description:**

Following his popular *The Hundred-Page Machine Learning Book*, Andriy Burkov delivers a highly concentrated engineering blueprint for building ML products. The book details project scoping, data collection, data preparation, feature engineering, model training,

hyperparameter optimization, model evaluation, deployment (REST APIs, cloud, embedded), post-deployment monitoring, and long-term system maintenance.

- **Key Features:**

- Direct, fluff-free, highly actionable rules for engineering execution.
- Real-world advice on handling imbalanced datasets, data leakage, and feature scaling.
- Concrete guidance on model serving, shadow deployments, A/B testing, and rollback strategies.

- **Technical Specifications:**

- **Author:** Andriy Burkov
- **Edition:** 1st Edition (2020)
- **Page Count:** ~290 pages
- **Language:** English

- **Package Contents:** 1x Paperback / Hardcover / eBook Edition.

- **Warranty:** Standard retailer return policy.

- **Shipping Information:** Dimensions: 6.0 x 0.7 x 9.0 inches; Weight: ~0.50 kg.

- **Frequently Asked Questions:**

- **Q:** *Is this book heavily math-focused?* **A:** No, it concentrates on practical engineering choices, project management, and execution tactics.

- **Customer Reviews:**

- "Zero filler. High-density practical advice on every single page."
- "The perfect practical field guide for practicing ML engineers."

6. Deep Learning with Python

- **Product ID:** 6

- **Product Name:** Deep Learning with Python (Second Edition)

- **Brand:** Manning Publications

- **Category:** Deep Learning / Computer Science / Programming

- **Price:** ~\$49.99 USD / ₹2,500 – ₹4,500 INR

- **Availability:** In Stock across all standard retailers.

- **Rating:** 4.8 / 5 stars (from 900+ global ratings)

- **Short Description:** An intuitive guide to deep learning written by the original creator of the Keras library.

- **Detailed Description:**

François Chollet introduces deep learning through clear, intuitive explanations without heavy mathematical abstractions. The second edition covers modern Keras and TensorFlow 2 developments, walking through core neural building blocks, computer vision models (CNNs, residual connections), sequence processing (RNNs, Transformers), text generation, generative models (VAEs, GANs), and best practices for model training, tuning, and generalization.

- **Key Features:**

- Written by François Chollet, creator of Keras and senior AI engineer at Google.
- Highly accessible visual explanations of tensor operations, backpropagation, and gradient descent.
- Hands-on implementation of text classification, image generation, time-series forecasting, and model optimization.

- **Technical Specifications:**

- **Author:** François Chollet
- **Edition:** 2nd Edition (Published Dec 2021)
- **Page Count:** 504 pages
- **Language:** English
- **Tech Stack:** Python 3, TensorFlow 2.x, Keras.
- **Package Contents:** 1x Paperback Book + complimentary PDF/eBook download through Manning Publications.
- **Warranty:** Retailer return policy for printing defects.
- **Shipping Information:** Dimensions: 7.4 x 1.1 x 9.2 inches; Weight: ~0.90 kg.
- **Frequently Asked Questions:**
- *Q: Do I need strong calculus skills for this book?* A: No, concepts are explained visually and programmatically using Python snippets.
- **Customer Reviews:**
- "Chollet has a rare gift for explaining complex technical ideas simply and visually."
- "The best conceptual intro to Keras and TensorFlow available."

7. Natural Language Processing with Transformers

- **Product ID:** 7
- **Product Name:** Natural Language Processing with Transformers: Building Language Applications with Hugging Face (2nd Edition / Revised)
- **Brand:** O'Reilly Media
- **Category:** Natural Language Processing / Deep Learning / LLMs
- **Price:** ~\$50.00 – \$69.99 USD / ₹2,800 – ₹5,500 INR
- **Availability:** In Stock.
- **Rating:** 4.8 / 5 stars (from 500+ global ratings)
- **Short Description:** The definitive practical guide to fine-tuning, training, and deploying Transformer architectures using Hugging Face.
- **Detailed Description:**

Authored by core Hugging Face engineers (Lewis Tunstall, Leandro von Werra, and Thomas Wolf), this book teaches how to build state-of-the-art NLP models. It provides hands-on walkthroughs for key tasks: text classification, named entity recognition (NER), text summarization, question answering, multi-lingual transfer learning, and model optimization (quantization, ONNX Runtime, pruning). It deeply covers Hugging Face libraries including `transformers`, `datasets`, `tokenizers`, and `accelerate`.

- **Key Features:**
- Deep dive into Attention mechanisms, BERT, GPT, T5, and RoBERTa architectures.
- Fine-tuning pre-trained language models on custom domain datasets.
- Production optimization strategies for high throughput and low-latency inference.
- Written by the team that built and maintains the open-source Hugging Face ecosystem.
- **Technical Specifications:**
- **Authors:** Lewis Tunstall, Leandro von Werra, Thomas Wolf
- **Edition:** 2nd / Revised Edition (2022/2023)
- **Page Count:** ~400 pages
- **Language:** English
- **Tech Stack:** Python, PyTorch, Hugging Face (`transformers`, `datasets`, `hub`).
- **Package Contents:** 1x Paperback Book or eBook; official code GitHub repo.
- **Warranty:** 30-day publisher warranty against physical defects.

- **Shipping Information:** Dimensions: 7.0 x 0.8 x 9.2 inches; Weight: ~0.72 kg.
- **Frequently Asked Questions:**
 - *Q: Is this applicable to Large Language Models (LLMs)?* A: Yes, Transformer architectures form the foundation of modern LLMs like GPT-4 and Llama.
- **Customer Reviews:**
 - "If you work with text and Transformers, this is an absolute must-have."
 - "Clear, practical code examples directly from Hugging Face core engineers."

8. Building Machine Learning Powered Applications

- **Product ID:**8
- **Product Name:** Building Machine Learning Powered Applications: Going from Idea to Product
- **Brand:** O'Reilly Media
- **Category:** Applied Machine Learning / Product Development / Software Engineering
- **Price:** ~\$49.99 USD / ₹1,800 – ₹3,800 INR
- **Availability:** In Stock.
- **Rating:** 4.6 / 5 stars (from 250+ global ratings)
- **Short Description:** A practical guide to taking machine learning ideas from initial feasibility checks to a functional product.
- **Detailed Description:**

Emmanuel Ameisen outlines a pragmatic framework for productizing ML models. The book takes you through assessing task feasibility, setting up end-to-end baseline pipelines, iterating on data quality and feature engineering, performing error analysis, debugging models systematically, serving models via web microservices (APIs), and creating intuitive frontend interfaces.

- **Key Features:**
 - Focuses heavily on creating fast, simple baseline models before jumping to complex neural networks.
 - Error analysis workflows: identifying performance bottlenecks through data slicing.
 - Turning standalone models into web APIs (Flask/FastAPI) integrated with user interfaces.
- **Technical Specifications:**
 - **Author:** Emmanuel Ameisen
 - **Edition:** 1st Edition (2020)
 - **Page Count:** 260 pages
 - **Language:** English
 - **Tech Stack:** Python, Scikit-Learn, Flask, REST APIs.
- **Package Contents:** 1x Paperback Book or Digital Copy.
- **Warranty:** Standard retailer book exchange policy.
- **Shipping Information:** Dimensions: 7.0 x 0.6 x 9.2 inches; Weight: ~0.48 kg.
- **Frequently Asked Questions:**
 - *Q: Is this book useful for software engineers with little ML background?* A: Yes, it emphasizes product engineering workflows and practical API delivery.
- **Customer Reviews:**
 - "Great blueprint for actually shipping models that deliver user value."
 - "Saves months of trial and error when building real AI apps."

9. Generative AI with Python and TensorFlow 2

- **Product ID:** 9
- **Product Name:** Generative AI with Python and TensorFlow 2: Create images, text, and music with VAEs, GANs, LSTMs, Transformer models
- **Brand:** Packt Publishing
- **Category:** Generative AI / Deep Learning / Creative Computing
- **Price:** ~\$44.99 USD / ₹999 – ₹4,111 INR
- **Availability:** In Stock.
- **Rating:** 4.4 / 5 stars (from 150+ global ratings)
- **Short Description:** A hands-on guide to generating synthetic image, text, and music data using deep generative architectures in TensorFlow 2.
- **Detailed Description:**

This book provides practical code walk-throughs for building generative AI models. Authors Joseph Babcock and Raghav Bali guide you through building Autoencoders, Variational Autoencoders (VAEs), Generative Adversarial Networks (DCGAN, CycleGAN, StyleGAN), text generators using Recurrent Networks and Transformers, and neural audio synthesis models for generating music scores.

- **Key Features:**
 - Hands-on implementation of GAN variations (CycleGAN for image-to-image translation, StyleGAN).
 - Text generation pipelines using Transformers and sequence-to-sequence models.
 - Creative AI applications combining visual and audio synthesis.
- **Technical Specifications:**
 - **Authors:** Joseph Babcock, Raghav Bali
 - **Edition:** 1st Edition (2021)
 - **Page Count:** 488 pages
 - **Language:** English
 - **Tech Stack:** Python 3, TensorFlow 2.x, Keras, Music21, OpenCV.
- **Package Contents:** 1x Print/Digital Book; access to code repository on GitHub.
- **Warranty:** Publisher return policy.
- **Shipping Information:** Dimensions: 7.5 x 1.0 x 9.2 inches; Weight: ~0.82 kg.
- **Frequently Asked Questions:**
 - *Q: Does it cover modern Diffusion Models?* A: It focuses primarily on VAEs, GANs, and early Transformer generators in TensorFlow 2.
- **Customer Reviews:**
 - "Fun, code-heavy book for developers interested in creative AI applications."

10. Probabilistic Machine Learning: An Introduction

- **Product ID:** 10
- **Product Name:** Probabilistic Machine Learning: An Introduction (Adaptive Computation and Machine Learning series)
- **Brand:** MIT Press
- **Category:** Advanced Machine Learning / Applied Statistics / Bayesian AI
- **Price:** ~\$95.00 USD / ₹4,500 – ₹8,500 INR
- **Availability:** In Stock; Hardcover and free open-access web version.
- **Rating:** 4.8 / 5 stars (from 400+ global ratings)

- **Short Description:** A modern, rigorous introduction to machine learning through a unified probabilistic and Bayesian framework.
- **Detailed Description:**

Written by Google Research Scientist Kevin P. Murphy, this text provides a comprehensive treatment of modern machine learning centered on probability and uncertainty modeling. It covers univariate/multivariate probability distributions, decision theory, optimization, linear models, deep neural networks, graphical models, variational inference, and generative models. It explicitly bridges classical probabilistic models with modern deep learning representations using Python libraries like JAX and PyTorch.

- **Key Features:**
 - Unified Bayesian probability perspective across all algorithms.
 - Detailed coverage of uncertainty quantification, variational inference, and MCMC methods.
 - Companion Python code notebooks implemented using JAX and PyTorch.
 - End-of-chapter exercises and mathematical derivations.
- **Technical Specifications:**
 - **Author:** Kevin P. Murphy
 - **Edition:** 1st Edition (Published 2022)
 - **Page Count:** 864 pages
 - **Language:** English
 - **Publisher:** MIT Press
 - **Code Stack:** Python, JAX, PyTorch, Probabilistic Programming (NumPyro).
- **Package Contents:** 1x Hardcover Book; access to full online code repository and interactive notebooks.
- **Warranty:** MIT Press/Retailer return window.
- **Shipping Information:** Dimensions: 8.3 x 1.8 x 10.3 inches; Weight: ~1.85 kg.
- **Frequently Asked Questions:**
 - *Q: How does this differ from Murphy's 2012 book?* A: It is a complete rewrite split into two volumes (*An Introduction* and *Advanced Topics*), incorporating modern deep learning and JAX framework implementations.
- **Customer Reviews:**
 - "An absolute masterpiece of mathematical clarity and modern ML coverage."
 - "Sets the benchmark for probabilistic machine learning textbooks."

AI Hardware Devices

1. NVIDIA Jetson Orin Nano Developer Kit

- **Product ID:** 11
- **Product Name:** NVIDIA Jetson Orin Nano Developer Kit
- **Brand:** NVIDIA
- **Category:** Embedded Systems / Edge AI / Single Board Computers
- **Price:** ~\$499.00 USD / ₹42,000 – ₹52,000 INR
- **Availability:** In Stock across major electronics distributors (Arrow, Seeed Studio, SparkFun, Robu.in)
- **Rating:** 4.8 / 5 stars

- **Short Description:** A compact, low-power edge AI development kit delivering up to 40 TOPS of AI performance for entry-level autonomous machines and vision systems.
- **Detailed Description:** The NVIDIA Jetson Orin Nano Developer Kit sets a high standard for entry-level edge AI and robotics. Powered by NVIDIA Ampere architecture GPU with 1024 CUDA cores and Tensor Cores, it delivers up to 80x the performance of the original Jetson Nano. It is designed to run modern AI workloads, including deep neural networks, transformer models, object detection, and multi-camera vision processing at low power consumption (7W to 15W). Supported by the comprehensive NVIDIA JetPack SDK, it allows developers to deploy AI frameworks like PyTorch, TensorFlow, TensorRT, and DeepStream seamlessly.
- **Key Features:**
 - Up to 40 TOPS of AI compute performance in a compact form factor.
 - NVIDIA Ampere GPU architecture with 1024 CUDA cores and 32 Tensor Cores.
 - Dual M.2 Key M slots for fast NVMe storage expansion.
 - Supports multiple concurrent camera streams via MIPI CSI-2 connectors.
- **Technical Specifications:**
 - **GPU:** 1024-core NVIDIA Ampere GPU with 32 Tensor Cores (up to 625 MHz)
 - **CPU:** 6-core Arm Cortex-A78AE v8.2 64-bit CPU (1.5 GHz)
 - **Memory:** 8GB 128-bit LPDDR5 (68 GB/s bandwidth)
 - **Storage:** microSD slot + 2x M.2 Key M (PCIe Gen3 x4/x2)
 - **Power:** 7W – 15W configurable operating power
 - **Networking:** 1x Gigabit Ethernet, M.2 Key E for Wi-Fi/Bluetooth
- **Package Contents:**
 - 1x Jetson Orin Nano carrier board and compute module
 - 1x 19V DC Power Supply
 - 1x Quick Start Guide
- **Warranty:** 1-Year Limited Hardware Warranty from NVIDIA / Authorized Distributor.
- **Shipping Information:** Dimensions: 100 mm x 79 mm x 21 mm (board); Weight: ~0.45 kg.
- **Frequently Asked Questions:**
 - *Q: Does it come with onboard eMMC storage?* A: No, storage must be provided via a microSD card or NVMe SSD plugged into the M.2 slot.
 - *Q: Can it run large language models (LLMs)?* A: Yes, small quantized LLMs (e.g., Llama-3-8B with INT4 optimization) can run using TensorRT-LLM.
- **Customer Reviews:**
 - "Incredible jump in performance compared to the original Jetson Nano."
 - "Essential for building low-latency computer vision applications on drones and mobile robots."

2. NVIDIA Jetson Xavier NX Developer Kit

- **Product ID:** 12
- **Product Name:** NVIDIA Jetson Xavier NX Developer Kit
- **Brand:** NVIDIA
- **Category:** Embedded AI Systems / Edge Computing
- **Price:** ~\$399.00 – \$499.00 USD / ₹35,000 – ₹45,000 INR
- **Availability:** Limited Stock / Managed Life Cycle (Legacy Architecture)

- **Rating:** 4.7 / 5 stars
- **Short Description:** A high-performance, power-efficient embedded system delivering up to 21 TOPS of AI compute in a credit-card-sized form factor.
- **Detailed Description:** The NVIDIA Jetson Xavier NX Developer Kit brings supercomputer-like performance to small edge devices. Built on the NVIDIA Volta GPU architecture, it offers up to 21 TOPS of accelerated computing at 10W to 15W. It is capable of running complex multi-modal AI pipelines, parallel neural networks, and high-resolution sensor processing simultaneously. Ideal for medical devices, commercial drones, industrial inspection, and smart city infrastructure.
- **Key Features:**
 - NVIDIA Volta GPU architecture with 384 CUDA cores and 48 Tensor Cores.
 - Dual NVDLA (NVIDIA Deep Learning Accelerator) engines for dedicated inference workloads.
 - Native support for NVIDIA JetPack, CUDA, TensorRT, and DeepStream SDKs.
 - Comprehensive I/O suite including HDMI, DisplayPort, Gigabit Ethernet, and MIPI CSI-2.
- **Technical Specifications:**
 - **GPU:** 384-core NVIDIA Volta GPU with 48 Tensor Cores
 - **CPU:** 6-core NVIDIA Carmel ARMv8.2 64-bit CPU
 - **Memory:** 8GB 128-bit LPDDR4x (51.2 GB/s)
 - **AI Performance:** Up to 21 TOPS (INT8)
 - **Storage:** microSD card slot / M.2 Key M NVMe slot
 - **Power:** Configurable 10W / 15W operational modes
- **Package Contents:**
 - 1x Jetson Xavier NX module and carrier board
 - 1x Power Adapter with regional cables
 - 1x Setup Guide
- **Warranty:** 1-Year Limited Manufacturer Warranty.
- **Shipping Information:** Dimensions: 103 mm x 90 mm x 31 mm; Weight: ~0.52 kg.
- **Frequently Asked Questions:**
 - *Q: What is the main difference between Xavier NX and Orin Nano? A:* Orin Nano uses the newer Ampere GPU architecture yielding up to 40 TOPS vs Xavier NX's 21 TOPS.
- **Customer Reviews:**
 - "Workhorse board for industrial AI vision applications."
 - "Extremely reliable for continuous video stream inference."

3. NVIDIA Jetson AGX Orin Developer Kit

- **Product ID:** 13
- **Product Name:** NVIDIA Jetson AGX Orin Developer Kit
- **Brand:** NVIDIA
- **Category:** Enterprise Edge AI / Autonomous Systems / Supercomputing
- **Price:** ~\$1,999.00 USD / ₹1,80,000 – ₹2,10,000 INR
- **Availability:** In Stock at specialized industrial and AI hardware suppliers.
- **Rating:** 4.9 / 5 stars
- **Short Description:** The ultimate edge AI developer kit delivering up to 275 TOPS of server-grade AI performance for advanced robotics and autonomous machinery.

- **Detailed Description:** The NVIDIA Jetson AGX Orin Developer Kit is NVIDIA's most powerful edge AI system, providing up to 275 TOPS of computing capability (over 8x the performance of Jetson AGX Xavier). Featuring an Ampere GPU with 2048 CUDA cores, 64 Tensor Cores, and a 12-core Arm Cortex-A78AE CPU, it allows developers to test and deploy complex multi-agent systems, real-time autonomous navigation, advanced generative AI models, and vision-language models directly at the edge.
- **Key Features:**
 - Extreme AI performance: Up to 275 TOPS (INT8) in a power budget of 15W to 60W.
 - 2048-core NVIDIA Ampere GPU + 2x NVDLA 2.0 deep learning engines.
 - Massive memory capacity: 64GB 256-bit LPDDR5 with 204.8 GB/s memory bandwidth.
 - Enterprise-grade expansion: 10GbE Ethernet, PCIe Gen4 expansion, and multi-camera support.
- **Technical Specifications:**
 - **GPU:** 2048-core Ampere GPU with 64 Tensor Cores (up to 1.3 GHz)
 - **CPU:** 12-core Arm Cortex-A78AE v8.2 64-bit CPU (up to 2.2 GHz)
 - **Memory:** 64GB LPDDR5
 - **Storage:** 64GB eMMC 5.1 onboard + M.2 Key M slot for NVMe
 - **Power:** Configurable 15W / 30W / 50W / 60W
 - **Networking:** 10 Gigabit Ethernet (RJ45)
- **Package Contents:**
 - 1x Jetson AGX Orin module with active heatsink/fan and carrier board
 - 1x 19V / 240W Power Adapter + Type-C cable
 - 1x USB-C to USB-A cable + Quick Start documentation
- **Warranty:** 1-Year Limited Manufacturer Warranty.
- **Shipping Information:** Dimensions: 110 mm x 110 mm x 71.6 mm; Weight: ~1.80 kg.
- **Frequently Asked Questions:**
 - *Q: Can this run full desktop generative AI and LLMs locally?* A: Yes, its 64GB unified memory allows running full 7B/13B parameter LLMs and Stable Diffusion pipelines locally at low latency.
- **Customer Reviews:**
 - "A server-class workstation in a desktop developer kit form factor."
 - "Unmatched performance for autonomous vehicle and heavy robotics prototyping."

4. Raspberry Pi 5 (8GB)

- **Product ID:** 14
- **Product Name:** Raspberry Pi 5 Model B (8GB RAM)
- **Brand:** Raspberry Pi Foundation
- **Category:** Single Board Computers / Hobbyist Electronics / IoT
- **Price:** ~\$80.00 USD / ₹7,200 – ₹8,900 INR
- **Availability:** In Stock across official Raspberry Pi Approved Resellers.
- **Rating:** 4.8 / 5 stars
- **Short Description:** A flagship single-board computer featuring a 64-bit quad-core Arm Cortex-A76 CPU, PCIe 2.0 interface, dual 4K display outputs, and 8GB RAM.

- **Detailed Description:** The Raspberry Pi 5 delivers a 2x to 3x increase in CPU and GPU performance over Raspberry Pi 4. Powered by the Broadcom BCM2712 quad-core 64-bit Arm Cortex-A76 processor running at 2.4 GHz, it features VideoCore VII graphics, dual 4K60 HDMI outputs, and custom RP1 I/O controller architecture. It includes dedicated power buttons, a real-time clock (RTC), a single-lane PCIe 2.0 interface for high-speed M.2 NVMe SSD expansion, and accelerated edge AI inference compatibility when paired with AI HAT extensions.
- **Key Features:**
 - 2.4 GHz Quad-core 64-bit Arm Cortex-A76 CPU with cryptography extensions.
 - VideoCore VII GPU supporting OpenGL ES 3.1 and Vulkan 1.2.
 - Onboard PCIe 2.0 x1 interface for external NVMe SSDs or AI accelerators.
 - Dual-band 802.11ac Wi-Fi, Bluetooth 5.0 / BLE, and Gigabit Ethernet.
- **Technical Specifications:**
 - **CPU:** Broadcom BCM2712 Quad-Core Arm Cortex-A76 @ 2.4 GHz
 - **RAM:** 8GB LPDDR4X-4267
 - **GPU:** VideoCore VII @ 800 MHz
 - **Display:** 2x micro-HDMI ports (up to dual 4K60p)
 - **Connectivity:** Gigabit Ethernet, 2x USB 3.0 (5Gbps), 2x USB 2.0, PCIe 2.0
 - **Power:** 5V / 5A DC via USB-C (Power Delivery compatible)
- **Package Contents:** 1x Raspberry Pi 5 (8GB) board only.
- **Warranty:** 1-Year Limited Standard Warranty.
- **Shipping Information:** Dimensions: 85 mm x 56 mm x 17 mm; Weight: ~0.08 kg.
- **Frequently Asked Questions:**
 - *Q: Does it require active cooling?* A: For heavy continuous workloads, an active cooler or fan case is strongly recommended to prevent thermal throttling.
 - *Q: Can it boot directly from an NVMe SSD?* A: Yes, via an official Raspberry Pi M.2 HAT or compatible PCIe expansion board.
- **Customer Reviews:**
 - "Feels like a full desktop replacement for everyday tasks."
 - "PCIe support makes storage performance lightning fast."

5. Google Coral Dev Board

- **Product ID:** 15
- **Product Name:** Google Coral Dev Board
- **Brand:** Google
- **Category:** Edge AI / Single Board Computers / Neural Processing
- **Price:** ~\$129.99 USD / ₹11,500 – ₹14,500 INR
- **Availability:** In Stock.
- **Rating:** 4.5 / 5 stars
- **Short Description:** A single-board computer with a removable System-on-Module (SoM) featuring Google's Edge TPU for ultra-fast TensorFlow Lite model inference.
- **Detailed Description:** The Google Coral Dev Board is a single-board computer designed for rapid prototyping of on-device AI applications. It combines an NXP i.MX 8M SoC with Google's proprietary **Edge TPU** coprocessor, capable of performing 4 trillion operations per second (4 TOPS) using only 2 Watts of power. It allows developers to run low-latency vision and audio deep learning models locally using TensorFlow Lite without cloud connectivity.

- **Key Features:**
 - Integrated Google Edge TPU delivering 4 TOPS (INT8) at 2W efficiency.
 - Modular design: Removable System-on-Module (SoM) containing CPU, RAM, eMMC, and Edge TPU.
 - Native integration with TensorFlow Lite and Google Coral Compiler.
 - Complete suite of hardware connections: HDMI, MIPI-CSI, Gigabit Ethernet, Wi-Fi, and Bluetooth.
- **Technical Specifications:**
 - **SoC:** NXP i.MX 8M (Quad-core Arm Cortex-A53 + Cortex-M4F)
 - **Co-processor:** Google Edge TPU (4 TOPS @ 8-bit)
 - **RAM:** 1GB, 2GB, or 4GB LPDDR4 options
 - **Storage:** 8GB eMMC onboard + microSD slot
 - **Connectivity:** Wi-Fi 5 (802.11ac), Bluetooth 4.2, Gigabit Ethernet
 - **Power:** 5V DC via USB Type-C
- **Package Contents:**
 - 1x Google Coral Dev Board (SoM + Baseboard)
- **Warranty:** 1-Year Hardware Warranty.
- **Shipping Information:** Dimensions: 88 mm x 60 mm x 24 mm; Weight: ~0.12 kg.
- **Frequently Asked Questions:**
 - *Q: What machine learning frameworks are supported?* A: Models must be compiled specifically for the Edge TPU using TensorFlow Lite (INT8 quantized models).
- **Customer Reviews:**
 - "Extremely low latency for live camera object detection."
 - "Super power-efficient for remote IoT deployments."

6. Google Coral USB Accelerator

- **Product ID:** 16
- **Product Name:** Google Coral USB Accelerator
- **Brand:** Google
- **Category:** AI Accelerators / USB Co-processors
- **Price:** ~\$59.99 – \$74.99 USD / ₹6,500 – ₹8,500 INR
- **Availability:** In Stock.
- **Rating:** 4.7 / 5 stars
- **Short Description:** A compact USB accessory that brings Edge TPU neural network coprocessing capabilities to existing Linux, Mac, and Windows systems.
- **Detailed Description:** The Google Coral USB Accelerator is a plug-and-play USB 3.0 device that integrates an Edge TPU coprocessor into host systems. It brings high-performance machine learning inference to devices like Raspberry Pi, industrial x86 PCs, or laptops. By offloading neural network math (4 TOPS at 2W), it allows host machines to execute high-frame-rate computer vision models (MobileNet, YOLO-Lite, PoseNet) locally without burdening host CPU resources.
- **Key Features:**
 - Google Edge TPU coprocessor (4 TOPS @ 8-bit INT8).
 - Standard USB 3.0 Type-C interface for high-speed host communication.
 - Cross-platform support: Debian/Ubuntu Linux, macOS, Windows 10/11, and Raspberry Pi OS.
 - Lightweight, portable, anodized aluminum casing.
- **Technical Specifications:**

- **ASIC:** Google Edge TPU
- **Performance:** 4 TOPS (INT8)
- **Interface:** USB 3.0 Type-C (Data + Power)
- **Supported Framework:** TensorFlow Lite
- **Operating Voltage:** 5V DC via USB port
- **Package Contents:**
 - 1x Google Coral USB Accelerator
 - 1x USB Type-C to Type-A Cable (30 cm)
- **Warranty:** 1-Year Manufacturer Warranty.
- **Shipping Information:** Dimensions: 65 mm x 30 mm x 8 mm; Weight: ~0.03 kg.
- **Frequently Asked Questions:**
 - *Q: Can I use PyTorch models with this? A:* PyTorch models must first be converted/quantized into TensorFlow Lite (.tflite) format before compiling for Edge TPU.
- **Customer Reviews:**
 - "Transformed my Raspberry Pi 4 into a real-time smart security camera processor."
 - "Small, affordable, and incredibly effective for object detection."

7. BeagleBone AI-64

- **Product ID:** 17
- **Product Name:** BeagleBone AI-64
- **Brand:** BeagleBoard.org
- **Category:** Industrial Single Board Computers / Open Source Hardware
- **Price:** ~\$119.00 USD / ₹11,000 – ₹13,500 INR
- **Availability:** In Stock.
- **Rating:** 4.4 / 5 stars
- **Short Description:** A powerful 64-bit open-source board built on Texas Instruments TDA4VM SoC delivering up to 8 TOPS for robotics and industrial edge computing.
- **Detailed Description:** BeagleBone AI-64 brings complete open-source hardware to industrial artificial intelligence. Powered by the Texas Instruments TDA4VM System-on-Chip featuring dual 64-bit Arm Cortex-A72 processors, C7x/C66x DSPs, and a dedicated Matrix Multiply Accelerator (MMA) pushing up to 8 TOPS. Designed with traditional BeagleBone cape expansion headers, mini DisplayPort, dual camera interfaces, and gigabit Ethernet, it is tailored for mobile robotics, autonomous navigation, and industrial automation.
- **Key Features:**
 - Dual-core Arm Cortex-A72 microprocessor subsystem @ 2.0 GHz.
 - Embedded Matrix Multiply Accelerator (MMA) delivering 8 TOPS for deep learning.
 - 4GB LPDDR4 RAM and 16GB onboard eMMC flash storage.
 - Unrivalled hardware expansion via standard BeagleBone P8/P9 Cape headers.
- **Technical Specifications:**
 - **Processor:** TI TDA4VM dual Cortex-A72 + 6x Cortex-R5F cores
 - **AI Accelerators:** C71x Vector DSP + MMA (8 TOPS @ INT8)
 - **RAM:** 4GB LPDDR4
 - **Storage:** 16GB eMMC 5.1 + MicroSD slot
 - **Video Output:** miniDisplayPort + DSI
 - **I/O:** 2x USB 3.0 Type-A, USB Type-C, Gigabit Ethernet, PCIe M.2 E-Key

- **Package Contents:**
 - 1x BeagleBone AI-64 development board
 - 1x Quick start instruction card
- **Warranty:** 1-Year Manufacturer Exchange Guarantee.
- **Shipping Information:** Dimensions: 104 mm x 83 mm x 23 mm; Weight: ~0.19 kg.
- **Frequently Asked Questions:**
 - *Q: Is software open source?* A: Yes, BeagleBoard.org provides a fully open-source Debian Linux environment and TI Edge AI toolchains.
- **Customer Reviews:**
 - "Fantastic open-source option for industrial robotics projects requiring real-time control."

8. Arduino Portenta X8

- **Product ID:** 18
- **Product Name:** Arduino Portenta X8
- **Brand:** Arduino
- **Category:** Industrial IoT / Hybrid SOM Microcontrollers
- **Price:** ~\$239.00 USD / ₹21,000 – ₹25,000 INR
- **Availability:** In Stock via Arduino Store and authorized distributors.
- **Rating:** 4.6 / 5 stars
- **Short Description:** A high-performance SOM board featuring 9 processing cores combining Linux-capable NXP i.MX 8M Mini with STM32H7 real-time microcontroller precision.
- **Detailed Description:** The Arduino Portenta X8 is an industrial-grade System-on-Module designed for enterprise IoT and edge computing. It combines a Linux application processor (NXP i.MX 8M Mini with 4x Cortex-A53 and 1x Cortex-M4) with an onboard real-time microcontroller (STM32H747XI dual Cortex-M7/M4). Preloaded with Linux OS and managed via Foundries.io cloud services, it can run containerized microservices (Docker) alongside deterministic real-time control loops, providing enterprise-level cybersecurity (PSA Certified).
- **Key Features:**
 - 9 Total Cores: 4x Cortex-A53, 2x Cortex-M4, 1x Cortex-M7, and dedicated cryptographic units.
 - Native Docker container runtime environment on Linux.
 - Dual High-Density connectors compatible with Portenta breakout boards.
 - Onboard Wi-Fi 5, Bluetooth 5.0, and hardware NXP SE050 Secure Element.
- **Technical Specifications:**
 - **Application Processor:** NXP i.MX 8M Mini (4x Arm Cortex-A53 @ 1.8 GHz + 1x Cortex-M4)
 - **Microcontroller:** STMicroelectronics STM32H747XI (Cortex-M7 @ 480 MHz + Cortex-M4 @ 240 MHz)
 - **RAM:** 2GB LPDDR4
 - **Storage:** 16GB eMMC Flash
 - **Security:** NXP SE050C2 Crypto chip + EdgeLock
 - **Form Factor:** Compact 66 mm x 25 mm SOM format
- **Package Contents:**
 - 1x Arduino Portenta X8 board
 - 1x Wi-Fi/Bluetooth antenna
- **Warranty:** 2-Year Official Arduino Warranty.

- **Shipping Information:** Dimensions: 66 mm x 25 mm; Weight: ~0.04 kg.
- **Frequently Asked Questions:**
 - *Q: Can I run Python or Node.js on Portenta X8?* A: Yes, full Linux OS support allows running Python, Node.js, Go, and Docker containers seamlessly.
- **Customer Reviews:**
 - "The ultimate fusion of real-time MCU control with high-level Linux cloud software."

9. STM32N6 Discovery Kit

- **Product ID:** 19
- **Product Name:** STM32N6 Discovery Kit (STM32N6570-DK)
- **Brand:** STMicroelectronics
- **Category:** Microcontrollers / Ultra-Low-Power Edge AI / Embedded Systems
- **Price:** ~\$223.00 USD / ₹19,000 – ₹22,000 INR
- **Availability:** In Stock via DigiKey, Mouser, and ST distributors.
- **Rating:** 4.7 / 5 stars
- **Short Description:** An advanced development platform for ST's first NPU-integrated microcontroller (STM32N6), delivering ultra-low-power computer vision and edge AI.
- **Detailed Description:** The STM32N6570-DK Discovery Kit showcases STMicroelectronics' flagship STM32N6 series microcontroller. Powered by an Arm Cortex-M55 core running at 800 MHz with Helium vector processing, it integrates the **ST Neural-ART Accelerator™** (an in-house NPU running at 1 GHz delivering 600 GOPS). Equipped with a 5-inch capacitive touchscreen LCD, dedicated ISP camera interface, H.264 video encoder, and 4.2 MB contiguous SRAM, it brings high-frame-rate computer vision and audio AI models directly to microcontroller power levels.
- **Key Features:**
 - STM32N657X0 microcontroller with Arm Cortex-M55 @ 800 MHz + Helium DSP extensions.
 - Proprietary ST Neural-ART Accelerator™ delivering efficient AI inference.
 - 5" 800x480 LCD display module with capacitive touch panel.
 - Onboard hardware ISP, MIPI CSI-2 camera connector, and H.264 video encoder.
- **Technical Specifications:**
 - **MCU:** STM32N657X0 (Cortex-M55 @ 800 MHz)
 - **NPU:** ST Neural-ART Accelerator (600 GOPS @ 1 GHz)
 - **SRAM:** 4.2 MB contiguous SRAM
 - **Flash / External RAM:** 1-Gbit Octo-SPI Flash + 256-Mbit PSRAM onboard
 - **Onboard Debugger:** STLINK-V3EC
 - **Expansion:** Arduino Uno R3, STMod+, MEMS audio connectors
- **Package Contents:**
 - 1x STM32N6570-DK board with integrated 5" LCD screen
 - 1x Onboard camera interface daughterboard / documentation
- **Warranty:** 1-Year Manufacturer Warranty.
- **Shipping Information:** Dimensions: ~150 mm x 100 mm x 25 mm; Weight: ~0.35 kg.
- **Frequently Asked Questions:**

- *Q: What software toolchain does it use?* A: Supported by STM32Cube.AI, Edge Impulse, STM32CubeIDE, and Keil MDK.
- **Customer Reviews:**
 - "Game changer for low-power vision on microcontrollers without needing Linux power budgets."

10. Intel Neural Compute Stick 2

- **Product ID:** 20
- **Product Name:** Intel Neural Compute Stick 2 (NCS2)
- **Brand:** Intel
- **Category:** USB Accelerators / Computer Vision / Edge AI
- **Price:** ~\$79.00 – \$99.00 USD / ₹7,500 – ₹9,800 INR
- **Availability:** Legacy / Secondary Retail Stock
- **Rating:** 4.4 / 5 stars
- **Short Description:** A fanless USB thumb-drive accelerator powered by the Intel Movidius Myriad X VPU for executing deep neural network inference on edge devices.
- **Detailed Description:** The Intel Neural Compute Stick 2 (NCS2) is designed to accelerate computer vision and deep learning inference at low power. Driven by the Intel Movidius Myriad X Vision Processing Unit (VPU), it features 16 SHAVE vector cores and a dedicated hardware neural network block. Plugging into any host x86 or ARM Linux system via USB 3.0, it integrates with the Intel OpenVINO Toolkit to optimize and execute models like CNNs for object detection, face recognition, and gesture control.
- **Key Features:**
 - Intel Movidius Myriad X VPU with 16 programmable SHAVE vector cores.
 - Hardware neural network accelerator yielding up to 1 TOPS of dedicated compute.
 - Standard USB 3.0 Type-A connector interface.
 - Seamless model optimization via Intel OpenVINO Toolkit SDK.
- **Technical Specifications:**
 - **VPU:** Intel Movidius Myriad X VPU
 - **Memory:** On-chip ultra-high bandwidth SRAM
 - **Interface:** USB 3.0 Type-A
 - **Supported Frameworks:** Caffe, TensorFlow, ONNX (converted via OpenVINO)
 - **Supported OS:** Ubuntu, Raspbian / Raspberry Pi OS, Windows 10
- **Package Contents:**
 - 1x Intel Neural Compute Stick 2
- **Warranty:** Standard 1-Year Limited Warranty.
- **Shipping Information:** Dimensions: 72.5 mm x 27 mm x 14 mm; Weight: ~0.04 kg.
- **Frequently Asked Questions:**
 - *Q: Can I plug multiple NCS2 sticks into one USB hub to multiply performance?* A: Yes, OpenVINO supports multi-device load balancing across multiple connected NCS2 sticks.
- **Customer Reviews:**
 - "Classic tool for accelerating vision inference on lightweight industrial PCs."

1. Apple HomePod (2nd Gen)

- **Product ID:** 21
- **Product Name:** Apple HomePod (2nd Generation)
- **Brand:** Apple
- **Category:** Smart Speakers / Smart Home Hubs / Audio Electronics
- **Price:** ~\$299.00 USD / ₹32,900 INR
- **Availability:** In Stock across Apple Stores, official online storefronts, and authorized Apple resellers.
- **Rating:** 4.7 / 5 stars
- **Short Description:** A premium high-fidelity smart speaker powered by Apple silicon, delivering computational audio, Spatial Audio with Dolby Atmos, and deep Siri integration.
- **Detailed Description:**

The 2nd Generation Apple HomePod combines advanced computational audio with a refined mesh design. Powered by the Apple S7 chip and system-sensing software, it continuously tunes room acoustics in real time. It features a custom 4-inch high-excursion woofer and a 5-tweeter array with directional beamforming for deep bass and clear vocal separation. Beyond music, it operates as a Matter-enabled smart home hub, features temperature and humidity sensors, and supports Sound Recognition for smoke and carbon monoxide alarms.

- **Key Features:**
 - High-excursion woofer with 20mm diaphragm and 5 beamforming tweeters.
 - Real-time room sensing for optimized acoustic reflection and Spatial Audio with Dolby Atmos.
 - Built-in temperature and humidity sensors for home automation triggers.
 - Ultra Wideband (UWB) proximity handoff for Seamless iPhone audio transfer.
- **Technical Specifications:**
 - **AI Assistant:** Siri (supports up to 6 multi-user voices)
 - **Processor:** Apple S7 Chip
 - **Connectivity:** Wi-Fi (802.11n), Bluetooth 5.0, Thread, Matter Hub
 - **Microphones:** 4-microphone array for far-field Siri
 - **Colors:** Midnight, White
- **Package Contents:**
 - 1x Apple HomePod (2nd Gen)
 - 1x Color-matched woven power cable
 - 1x Quick Start Documentation
- **Warranty:** 1-Year Limited Manufacturer Warranty (AppleCare+ eligible).
- **Shipping Information:** Dimensions: 168 mm x 142 mm; Weight: ~2.3 kg.
- **Frequently Asked Questions:**
 - *Q: Can I pair this with a 1st Gen HomePod for stereo sound?* A: No, stereo pairing requires two identical generation HomePod speakers.

- *Q: Does it work with Android devices?* A: No, full setup and functionality require an iOS or iPadOS device running current software.
- **Customer Reviews:**
 - "Sound quality is unmatched for its size, especially when streaming Dolby Atmos tracks via Apple Music."

2. Amazon Echo (4th Gen)

- **Product ID:** 22
- **Product Name:** Amazon Echo (4th Generation)
- **Brand:** Amazon
- **Category:** Smart Speakers / Smart Home Hubs
- **Price:** ~\$99.99 USD / ₹9,999 INR
- **Availability:** In Stock across Amazon online platforms and electronics retailers.
- **Rating:** 4.7 / 5 stars
- **Short Description:** A spherical smart speaker with premium sound, a built-in Zigbee smart home hub, and hands-free Alexa control.
- **Detailed Description:**

The 4th Generation Amazon Echo combines rich, detailed sound with an integrated smart home hub. Driven by Amazon's AZ1 Neural Edge processor for faster response processing, it features a 3.0-inch neodymium woofer and dual 0.8-inch front-firing tweeters that automatically adapt to room acoustics. Its built-in Zigbee, Matter, and Sidewalk protocols allow seamless local connection and control of compatible smart lights, locks, and sensors without extra bridge hardware.

- **Key Features:**
 - Spherical design with dynamic sound output (Dolby Audio processing).
 - Built-in Zigbee & Matter smart home hub for local device control.
 - AZ1 Neural Edge processor for rapid local speech processing.
 - Multi-room music synchronization across compatible Echo devices.
- **Technical Specifications:**
 - **AI Assistant:** Amazon Alexa
 - **Speakers:** 3.0" (76.2 mm) woofer and two 0.8" (20 mm) tweeters
 - **Connectivity:** Dual-band Wi-Fi (802.11a/b/g/n/ac), Bluetooth (A2DP), Zigbee, Matter, Amazon Sidewalk
 - **Ports:** 3.5 mm line in/out line audio jack
 - **Colors:** Charcoal, Glacier White, Twilight Blue
- **Package Contents:**
 - 1x Amazon Echo (4th Gen)
 - 1x 30W Power Adapter
 - 1x Quick Start Guide
- **Warranty:** 1-Year Limited Manufacturer Warranty.
- **Shipping Information:** Dimensions: 144 mm x 144 mm x 133 mm; Weight: ~970 g.
- **Frequently Asked Questions:**
 - *Q: Can I use it as an external speaker for my TV?* A: Yes, via 3.5mm line-in, Bluetooth, or wireless pairing with Fire TV devices.
- **Customer Reviews:**
 - "Huge audio upgrade over previous generations, and the built-in Zigbee hub saves money on smart light bridges."

3. Amazon Echo Dot (5th Gen)

- **Product ID:** 23
- **Product Name:** Amazon Echo Dot (5th Generation)
- **Brand:** Amazon
- **Category:** Compact Smart Speakers / Micro Assistants
- **Price:** ~\$49.99 USD / ₹5,499 INR
- **Availability:** In Stock globally.
- **Rating:** 4.7 / 5 stars
- **Short Description:** Amazon's most popular compact smart speaker featuring improved audio clarity, vibrant bass, and built-in climate motion sensors.
- **Detailed Description:**

The Echo Dot 5th Gen offers upgraded audio capabilities in a small, spherical form factor. It packs an enlarged 1.73-inch front-firing driver delivering clearer vocals and double the bass output compared to previous generations. Beyond voice requests with Alexa, it includes built-in indoor temperature sensors and accelerometer motion detection to trigger automated routines (like turning on lights when entering a room or setting climate rules).

- **Key Features:**
 - Redesigned 1.73-inch speaker driver for enhanced bass and vocal clarity.
 - Integrated indoor temperature sensor and ultrasound motion detection.
 - Tap gesture controls (pause music, dismiss timers, snooze alarms).
 - Privacy controls including a hardware microphone-off button.
- **Technical Specifications:**
 - **AI Assistant:** Amazon Alexa
 - **Audio:** 1.73" (44 mm) front-firing speaker
 - **Connectivity:** Dual-band Wi-Fi 5, Bluetooth, Matter mesh support
 - **Sensors:** Temperature, Ultrasound motion, Accelerometer
 - **Colors:** Charcoal, Deep Sea Blue, Glacier White
- **Package Contents:**
 - 1x Amazon Echo Dot (5th Gen)
 - 1x 15W Power Adapter
 - 1x Quick Start Guide
- **Warranty:** 1-Year Limited Warranty.
- **Shipping Information:** Dimensions: 100 mm x 100 mm x 89 mm; Weight: ~304 g.
- **Frequently Asked Questions:**
 - *Q: Does this model have a clock display?* A: This standard version does not, but an "Echo Dot with Clock" variant exists separately.
- **Customer Reviews:**
 - "Incredible sound for such a small device. Temperature sensor integration makes smart thermostat routines effortless."

4. Amazon Echo Show 8 (3rd Gen)

- **Product ID:** 24
- **Product Name:** Amazon Echo Show 8 (3rd Generation)
- **Brand:** Amazon

- **Category:** Smart Displays / Smart Home Controls
- **Price:** ~\$149.99 USD / ₹14,999 INR
- **Availability:** In Stock.
- **Rating:** 4.6 / 5 stars
- **Short Description:** An 8-inch HD smart display featuring spatial audio, auto-framing 13MP camera, and built-in smart home hub.
- **Detailed Description:**

The 3rd Gen Echo Show 8 redefines the smart display experience with edge-to-edge glass and room-filling spatial audio. Driven by an updated processor delivering faster local response times, its 8-inch HD touchscreen displays adaptive dynamic home content depending on your proximity. Features a centered 13MP camera that auto-frames during video calls, as well as an integrated Zigbee, Thread, and Matter hub to manage smart home networks locally.

- **Key Features:**
 - 8.0-inch HD touchscreen display with adaptive color and proximity sensing.
 - Spatial audio processing with room calibration for immersive sound output.
 - 13MP camera with auto-framing technology and physical privacy shutter.
 - Built-in Zigbee, Thread, and Matter smart home hub.
- **Technical Specifications:**
 - **AI Assistant:** Amazon Alexa
 - **Display:** 8.0" Touchscreen (1280 x 800 resolution)
 - **Camera:** 13 MP with built-in physical shutter
 - **Audio:** 2x 2.0" neodymium drivers with passive bass radiator
 - **Connectivity:** Dual-band Wi-Fi, Bluetooth, Zigbee, Thread, Matter
- **Package Contents:**
 - 1x Amazon Echo Show 8 (3rd Gen)
 - 1x 30W Power Adapter + 1.5m Cable
 - 1x Quick Start Guide
- **Warranty:** 1-Year Limited Warranty.
- **Shipping Information:** Dimensions: 200 mm x 139 mm x 106 mm; Weight: ~1.03 kg.
- **Frequently Asked Questions:**
 - *Q: Can I stream video content like Prime Video, Netflix, or YouTube?* A: Yes, video streaming apps and built-in web browser options are natively supported.
- **Customer Reviews:**
 - "The ideal kitchen counter companion for recipes, video calling, and home security monitoring."

5. Google Nest Audio

- **Product ID:** 25
- **Product Name:** Google Nest Audio
- **Brand:** Google
- **Category:** Smart Speakers / Audio Systems
- **Price:** ~\$99.99 USD / ₹7,999 INR
- **Availability:** In Stock across Google Store and retail partners.
- **Rating:** 4.6 / 5 stars

- **Short Description:** An eco-designed, music-first smart speaker delivering 75% louder volume and 50% stronger bass than the original Google Home.
- **Detailed Description:**

Google Nest Audio is built from the ground up for high-fidelity music playback and intelligent voice support. Encased in 70% recycled fabric, it features a 75mm woofer and 19mm tweeter for clean vocals and deep bass. Media EQ automatically tunes the audio profile depending on what you're listening to (music, news, or podcasts), while Ambient IQ adjusts assistant volume based on background room noise.

- **Key Features:**
 - Dedicated 75mm woofer and 19mm tweeter tuned for acoustic accuracy.
 - Media EQ & Ambient IQ adaptive sound optimizations.
 - Built-in Google ML hardware engine for fast local assistant response times.
 - Stereo pairing and multi-room grouping via Google Home ecosystem.
- **Technical Specifications:**
 - **AI Assistant:** Google Assistant
 - **Processor:** Quad-core A53 @ 1.8 GHz with ML hardware engine
 - **Connectivity:** Wi-Fi 5 (802.11ac), Bluetooth 5.0, Chromecast built-in
 - **Microphones:** 3 far-field microphones with physical mute switch
 - **Colors:** Chalk, Charcoal, Sage, Sand, Sky
- **Package Contents:**
 - 1x Google Nest Audio
 - 1x 30W Power Adapter + 1.5m Cable
 - 1x Documentation Package
- **Warranty:** 1-Year Manufacturer Warranty.
- **Shipping Information:** Dimensions: 175 mm x 124 mm x 78 mm; Weight: ~1.2 kg.
- **Frequently Asked Questions:**
 - *Q: Can I stream directly from my phone without Wi-Fi?* A: Yes, it can function as a standard Bluetooth speaker once initial setup is complete.
- **Customer Reviews:**
 - "Tremendous sound upgrade for Google Assistant users; vocals and acoustic instruments sound crystal clear."

6. Google Nest Hub (2nd Gen)

- **Product ID:** 26
- **Product Name:** Google Nest Hub (2nd Generation)
- **Brand:** Google
- **Category:** Smart Displays / Nightside Sleep Assistants
- **Price:** ~\$99.99 USD / ₹7,999 – ₹8,999 INR
- **Availability:** In Stock.
- **Rating:** 4.5 / 5 stars
- **Short Description:** A 7-inch display smart home hub featuring contactless Soli radar Sleep Sensing, upgraded bass, and Google Assistant.
- **Detailed Description:**

The 2nd Generation Google Nest Hub serves as the central control dashboard for smart homes. Featuring a 7-inch frameless glass touchscreen with enhanced speaker hardware (delivering 50% more bass), its standout innovation is Google's **Soli radar**

technology. Placed on a bedside table, Sleep Sensing tracks movement and breathing patterns without needing wearable devices or a camera, providing detailed night-time sleep analysis.

- **Key Features:**
 - Soli radar-based contactless Sleep Sensing (no wearables or cameras needed).
 - 7-inch WSVGA touchscreen display with Ambient EQ brightness adjustment.
 - Enhanced speaker acoustics providing 50% more low-end bass response.
 - Motion Sense gesture controls (wave hand to pause music or snooze alarms).
- **Technical Specifications:**
 - **AI Assistant:** Google Assistant
 - **Display:** 7.0-inch Touchscreen (1024 x 600 resolution)
 - **Sensors:** Soli radar, Ambient EQ light sensor, Temperature sensor
 - **Connectivity:** Wi-Fi 5, Bluetooth 5.0, Chromecast built-in, Thread boundary router
 - **Camera:** None (designed for bedroom privacy)
- **Package Contents:**
 - 1x Google Nest Hub (2nd Gen)
 - 1x 15W Power Adapter
 - 1x Quick Start Guide
- **Warranty:** 1-Year Limited Hardware Warranty.
- **Shipping Information:** Dimensions: 177 mm x 120 mm x 69.5 mm; Weight: ~558 g.
- **Frequently Asked Questions:**
 - *Q: Does it have a camera?* A: No, it is camera-free for enhanced bedroom privacy, relying solely on Soli radar for gesture and sleep tracking.
- **Customer Reviews:**
 - "The sleep tracking insights are surprisingly accurate without having to wear a smart watch to bed."

7. Google Nest Mini

- **Product ID:** 27
- **Product Name:** Google Nest Mini (2nd Generation)
- **Brand:** Google
- **Category:** Micro Smart Speakers / Voice Assistants
- **Price:** ~\$49.00 USD / ₹4,499 INR
- **Availability:** In Stock globally.
- **Rating:** 4.5 / 5 stars
- **Short Description:** Google's iconic wall-mountable compact smart speaker featuring improved voice recognition and 2x stronger bass performance.
- **Detailed Description:**

The Google Nest Mini is an entry-level compact smart speaker engineered to fit anywhere in the home. Featuring a built-in wall mount latch, it packs double the bass power of the original Google Home Mini. Equipped with a dedicated machine learning chip on board, it processes common Google Assistant voice commands locally for faster response latency. Ultrasonic proximity sensors illuminate touch volume controls as your hand approaches.

- **Key Features:**

- 2x stronger bass response and refined mid-range audio output.
- On-device Machine Learning chip for faster command execution.
- Integrated wall mount latch on the underside casing.
- Ultrasonic proximity sensing for capacitive LED control lighting.
- **Technical Specifications:**
 - **AI Assistant:** Google Assistant
 - **Driver:** 40 mm driver with 360-degree sound
 - **Microphones:** 3 far-field microphones with physical mute slider
 - **Connectivity:** Wi-Fi 5, Bluetooth 5.0, Chromecast built-in
 - **Colors:** Chalk, Charcoal, Coral, Sky
- **Package Contents:**
 - 1x Google Nest Mini
 - 1x 15W DC Power Adapter + Cable
 - 1x Documentation Package
- **Warranty:** 1-Year Limited Warranty.
- **Shipping Information:** Dimensions: 98 mm x 98 mm x 42 mm; Weight: ~181 g.
- **Frequently Asked Questions:**
 - *Q: Can I mount this directly on a wall screw?* A: Yes, it features a custom integrated keyhole wall mount built directly into its rear housing.
- **Customer Reviews:**
 - "Perfect small assistant for setting kitchen timers, checking the weather, and streaming background music."

8. Apple HomePod Mini

- **Product ID:** 28
- **Product Name:** Apple HomePod Mini
- **Brand:** Apple
- **Category:** Compact Smart Speakers / Siri Hubs
- **Price:** ~\$99.00 USD / ₹9,900 INR
- **Availability:** In Stock across all Apple retail outlets and authorized distributors.
- **Rating:** 4.8 / 5 stars
- **Short Description:** A compact sphere delivering 360-degree computational audio, Siri intelligence, and Thread/Matter smart home connectivity.
- **Detailed Description:**

Standing just 3.3 inches tall, the Apple HomePod Mini delivers rich 360-degree audio output. Powered by the Apple S5 chip, it uses computational audio to analyze performance 180 times per second to maximize dynamic range and bass response. It acts as an Apple Home hub with native support for Thread networking and Matter protocol. Built-in temperature and humidity sensors trigger automated smart home actions, while Ultra Wideband technology enables physical tap-to-transfer audio handoffs.

- **Key Features:**
 - Full-range driver and dual passive radiators for 360-degree acoustic output.
 - Apple S5 chip running real-time computational audio processing algorithms.
 - Integrated Thread boundary router & Matter hub for smart home automation.
 - Proximity audio Handoff powered by Apple's U1 Ultra Wideband chip.
- **Technical Specifications:**

- **AI Assistant:** Siri (Multi-user voice recognition)
- **Sensors:** Temperature & Humidity sensors, Proximity sensor
- **Connectivity:** Wi-Fi (802.11n), Bluetooth 5.0, Thread, U1 Ultra Wideband
- **Colors:** Yellow, Orange, Blue, Midnight, White
- **Package Contents:**
 - 1x Apple HomePod Mini (with attached USB-C cable)
 - 1x 20W USB-C Power Adapter
 - 1x Quick Start Guide
- **Warranty:** 1-Year Limited Manufacturer Warranty (AppleCare+ eligible).
- **Shipping Information:** Dimensions: 84.3 mm x 97.9 mm; Weight: ~345 g.
- **Frequently Asked Questions:**
 - *Q: Can two HomePod Minis be paired together?* A: Yes, two HomePod Minis in the same room can be configured as a stereo pair.
- **Customer Reviews:**
 - "Astonishing audio output for its tiny size. Seamless integration with the Apple ecosystem."

9. Lenovo Smart Clock Essential (with Alexa)

- **Product ID:** 29
- **Product Name:** Lenovo Smart Clock Essential with Alexa Built-in
- **Brand:** Lenovo
- **Category:** Smart Clocks / Bedside Assistant Devices
- **Price:** ~\$49.99 USD / ₹3,999 – ₹4,999 INR
- **Availability:** In Stock.
- **Rating:** 4.3 / 5 stars
- **Short Description:** A smart bedside clock featuring a high-contrast LED display, built-in Alexa, ambient nightlight, and USB charging capabilities.
- **Detailed Description:**

The Lenovo Smart Clock Essential merges traditional bedside digital alarm clock simplicity with smart voice capability. Its 4-inch LED display presents clear time, weather, decibel measurements, and alarm details without intrusive video distractions. Equipped with Amazon Alexa built-in, it allows setting timers, managing routines, streaming audio through its 3W speaker, and controlling room lighting via voice commands.

- **Key Features:**
 - 4-inch high-visibility digital LED clock display showing weather and decibel status.
 - Integrated soft ambient nightlight built around the rear perimeter casing.
 - Built-in USB Type-A port on the back for overnight phone charging.
 - Physical alarm setup, volume, and microphone mute buttons.
- **Technical Specifications:**
 - **AI Assistant:** Amazon Alexa built-in
 - **Speaker:** 1.5-inch 3W full-range audio speaker (~92 dB max)
 - **Display:** 4-inch monochrome non-touch LED display
 - **Connectivity:** Wi-Fi 5 (802.11 b/g/n), Bluetooth 5.1
 - **Colors:** Misty Blue, Clay Grey
- **Package Contents:**

- 1x Lenovo Smart Clock Essential (Alexa)
- 1x 12V / 1.5A DC Power Adapter
- 1x Quick Setup Guide
- **Warranty:** 1-Year Limited Manufacturer Warranty.
- **Shipping Information:** Dimensions: 113.5 mm x 93.3 mm x 71.3 mm; Weight: ~249 g.
- **Frequently Asked Questions:**
 - *Q: Can I view camera feeds on the screen?* A: No, the display is a simple digital LED screen designed for clock digits and weather icons.
- **Customer Reviews:**
 - "The ideal nightstand clock. No camera or complex screen distraction, just clear digits and Alexa convenience."

10. JBL Authentics 500 Smart Speaker

- **Product ID:** 30
- **Product Name:** JBL Authentics 500 High-Fidelity Smart Speaker
- **Brand:** JBL
- **Category:** Premium Audio / Multi-Assistant Smart Speakers
- **Price:** ~\$699.99 USD / ₹59,999 INR
- **Availability:** In Stock across high-end audio outlets and JBL official platforms.
- **Rating:** 4.8 / 5 stars
- **Short Description:** A retro-inspired 270W 3.1-channel speaker featuring Dolby Atmos, concurrent Google Assistant and Alexa operation, and high-res Wi-Fi streaming.
- **Detailed Description:**

The JBL Authentics 500 combines 1970s classic heritage design (Quadrex grille, aluminum frame, leather-like enclosure) with cutting-edge audio architecture. Delivering 270 Watts of 3.1-channel sound with virtual Dolby Atmos, it features three 1-inch aluminum tweeters, three 2.75-inch midrange drivers, and a 6.5-inch down-firing subwoofer. It is engineered with simultaneous support for both Google Assistant and Amazon Alexa, allowing users to invoke either voice assistant on the same physical speaker dynamically.

- **Key Features:**
 - Simultaneous Voice Assistants: Google Assistant and Amazon Alexa run concurrently.
 - 270W 3.1-Channel acoustic output with immersive Dolby Atmos 3D Music rendering.
 - Automatic room self-tuning calibration every time the device powers on.
 - Multi-room streaming via AirPlay, Alexa MRM, Chromecast built-in, and Spotify Connect.
- **Technical Specifications:**
 - **AI Assistants:** Google Assistant & Amazon Alexa (Concurrent)
 - **Power Output:** 270 Watts Max
 - **Drivers:** 3x 1" tweeters + 3x 2.75" mid-range + 1x 6.5" down-firing woofer
 - **Connectivity:** Wi-Fi 6 (802.11ax), Bluetooth 5.3, Ethernet (RJ45), 3.5mm Aux in
 - **Dimensions:** 447 mm x 240 mm x 255 mm; Weight: ~7.8 kg

- **Package Contents:**
 - 1x JBL Authentics 500 Speaker
 - 1x Regional Power Cable
 - 1x Quick Start Guide & Safety Sheet
- **Warranty:** 1-Year Official JBL/Harman Warranty.
- **Shipping Information:** Weight: ~7.8 kg net; heavy-duty premium packaging.
- **Frequently Asked Questions:**
 - *Q: Do I have to pick between Google Assistant and Alexa?* A: No, both assistants can be registered and used interchangeably without switching settings.
- **Customer Reviews:**
 - "The best-sounding smart speaker on the market. Powerful bass response combined with stunning retro aesthetics."